PHY 1100: General Physics I

Fall 2025

Instructor Information

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Student Hours: M and W 1:00pm to 3:00pm. If none of these times work, send me an email so we can find a suitable time. Meetings can be in-person or via zoom. Link will be posted on Blackboard in the first week of class.

My student hours are protected time exclusively designated for my students to meet with me, chat about anything they might need, ask questions about course material and/or grading, or just to say hi.

Attention: Announcements made in class will *usually* be circulated over the Blackboard. Ensure you receive Blackboard announcement notifications in your email/through the Blackboard app.
Check your villanova.edu email regularly!

Class Meetings:

Lectures + Recitation

Section 9:	MW	4:00 pm – 5:15 pm	Mendel Science Center, Room 154
Section 4:	TTh	3:20 pm – 4:35pm	Mendel Science Center, Room 254

Each session will have a 50 min interactive lecture and a 25 min recitation session

You may not attend a different session from what you are registered for without prior approval of your instructor.

Catalog Description:

PHY 1100 - General Physics I (Mechanics, sound, and fluid dynamics. Recommended for Biology majors.)
3 Credits. **Prerequisites:** MAT 1310, 1312, 1320, 1400, or 1500 with a Minimum Grade of D-

Note that his syllabus is subject to change.

Overview/Course Description:

In this first semester of general physics, the main topics will be kinematics, mechanics, wave motion, and fluids. Calculus will be used for the derivation of certain relationships. Mathematical relationships between physical observables are expressed algebraically so that knowledge of calculus beyond the MAT 1310 or 1312 level is not required. Emphasis will be placed on the application of physical principles to the solution of practical problems. The methodology of problem solving will be stressed. The assigned problems represent a minimum of expected work. You are encouraged to try other problems. *It is assumed that students have a good working knowledge of algebra, trigonometry, geometry, and calculus.*

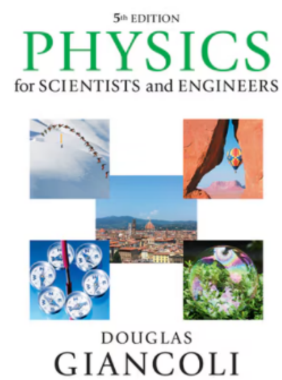
Course Learning Objectives and Academic Goals:

This course is a calculus/trigonometry-based course designed to deepen understanding of basic physical principles. The course focuses on physics of motion, mechanics, matter and energy. Emphasis is on conceptual development and systematic problem solving. *The analytical skills you develop in this course will have wide-ranging applications in your future professional development.*

After completion of this course, successful students will be able to

- Analyze natural processes from a broad physics perspective
- Explain how certain technology works and the principles that govern it
- Apply fundamental concepts from the covered units to solve problems
- Make physics-informed numerical estimates of interesting quantities
- Analyze the application of physics to real-life situations

Textbook:



"**Physics for Scientists & Engineers** " by Duglas Giancoli 5th edition

- ISBN 13: 9780321992277 (Student edition)
- ISBN 13: 9780134378060 (Classic Student edition)
- ISBN 13: 9780134378084 (Looseleaf edition)

Attendance and Responsibilities:

Regular attendance at lectures and recitations are essential if the student is to keep pace with the course. Students are encouraged to take notes and ask questions at any time during a regular lecture or recitation. They are urged to seek the instructor's help during the posted office hours, or at any other mutually convenient time. Physics is not an easy subject to learn, and *a few days of preparation just before a test will not be sufficient.* Students are expected to read the material in the textbook ahead of time and on a daily basis. It is virtually impossible to attain good grades by playing "catch-up physics" just prior to each test.

Your weekly "to-do" list

You should invest 7-10 hours per week in this class. Each week students are expected to:

- **Read the corresponding chapters** before coming to class.
- **Come to class ready to participate**, do not hesitate in asking questions at any stage of the lecture and/or recitation. If you are not ready, I still want you to come to class.
- **Take detailed notes and review these notes after class.**
- **Work out the assigned exercises, either on your own or as part of a study group**
- At any stage, please **come to student hours to ask questions (There are not stupid questions!) or just to say hi** and share

Learning and developing skills require time, dedication, and practice. My aim as your instructor is to help you develop problem solving skills and thinking processes that are transferable to any career path you wish to pursue. Thus, all the time we put into this class is an investment towards your success as a professional beyond letter grades.

My commitment

I strive for my classroom to be a welcoming and safe learning environment where students can express ideas, ask questions, interact with each other, brainstorm about problem-solving, make mistakes and learn from them. During our time together, we will perform a variety of hands-on learning activities. My goal is to be a coach of your learning process, help unstuck you when you feel stuck and/or confused, providing scaffolded steps for completing the activity and supporting you as you complete them. For me, it is of the highest importance that my students feel comfortable to be who they are (unapologetically). Society expectations on how scientists and scholars look, ask questions, speak a particular dialect, or articulate their ideas are mere social constructs. In my classroom, you are the scientists and scholars in training. Thus, **your ideas and questions – including both confusion and eureka moments -- are always valid. Note: "Huh?", "Say Whaaat?" and "Word?" are completely valid questions.**

Your rights in my classroom

When we are learning and aiming for excellence, we can also suffer from perfectionism and set unrealistic expectations for ourselves. Thus, I would like to remind you of some your inalienable rights as my student:

- Right to succeed
- Right to make mistakes
- Right to understand a concept or technique
- Right to struggle understanding a concept or technique
- Right to learn using a variety of tools and/or approaches
- Right to be respected
- Right to feel safe and secure
- Right to speak up
- Right to express your culture, background, religious belief, spiritual practice, national origin, race, gender identity and/or orientation, among other characteristics
- Right to be human

Unitas – Our Learning Community

We will be working together during the whole term. Just as we might find ourselves "in the zone" tackling concepts, sometimes we might find ourselves struggling or stuck. Remember to be mindful and considerate of your fellow classmates. We are all in the same team with the goals of learning and growing. In academia, we cannot state we are pursuing knowledge to the highest standard without including the perspectives of scholars from all races, national origin, backgrounds, socioeconomic circumstances, gender orientation, gender identities, religious beliefs and spiritual practices, including able-bodied and differently abled, providing them what they need to succeed. In the overall scientific community, we all still have work to do towards achieving this standard. Yet, it is my hope to provide a positive learning environment for everyone in our classroom as a microcosmos. If there is anything that I can do differently to better serve your needs, remember your "right to speak up" and let me know. I am happy to help in any way possible.

AI use

I have found AI to be unreliable for physics analysis and communicating science. Thus, for the sake of your learning process, you are not allowed to use AI. You might use it to check your grammar and/or spelling, if appropriate. Even then, take its suggested edits with a grain of salt and trust your intuition.

Class structure

Recitations:

Recitation time will be dedicated to problem solving skills and the administration of quizzes (see more detail on these below). You will be given an assignment of text problems that you will be expected to have given a good faith attempt at the problems before recitation. **There is no graded "homework" in the course. The "problem assignments" are for YOU!**

Exams:

There will be two exams given during the semester on the dates below. Each test is worth **100 points** and the material for each test is tentative. The final exam will be cumulative and will count **150 points**. For those who miss an exam for a ***legitimate reason*****,** the final exam will be scaled to count for the missed exam. **There will be no make-up exams.** See "Tentative Schedule" for anticipated exam dates. An equation sheet will be provided with each set of exam questions. Numerical values of constants and conversion factors will also be provided. The student must know the meaning of the symbols since they will not be defined on this supplementary sheet. No phones or headphones are allowed during the exam or quiz.

In grading the semester exams, particular attention will be given to the work done and the evidence of the thought behind it. An isolated numerical answer, even if correct or the mere transcription of a listed formula will not be credited.

The Final exam (cumulative) is scheduled on **Wednesday December 17th** 11:30am-1:30pm Location TBD.

For students with documented academic time conflict, please notify the instructor as soon as possible or at least three weeks prior to the exam date.
Alternative Final Exam date with notification of the instructor ONLY: TBD

Quizzes:

Quizzes are worth 100 points of your final grade (10 quizzes worth 10 points each). Quizzes will be given during the recitation. No phones, headphones, or restroom breaks are allowed during the quiz.

If our "Tentative Schedule" is altered enough that ten quizzes cannot be completed, the average of the quizzes that are completed will be scaled to 100 points.

We do not offer makeup quizzes. **If a quiz is missed** for a ***legitimate reason*****,** your average quiz **score** for the **semester** will be your missed quiz grade. This rule does not apply if a student misses more than one quiz in a row (for special cases please talk to the instructor) or to the last quiz of the semester (quiz #10).

Tutors:

A list of tutors will be available in the Physics office (M 347). This tutoring service will be free of charge as the department pays students who have previously excelled in the course to serve as tutors. One-on-one tutors will also be available at a reasonable rate.

Assigned Problems

On average, one set of problems per week will be assigned from the textbook. These assigned problems are designed to help students in understanding the material, **no grade** is assigned to the assigned HW.

Grades:

Your final grade is determined by the number of points you earn; the percent column is provided to help you interpret what the points mean. However, there *may* be a curve applied to the final grades which may lower the thresholds for each of the defined letter grades. That is, if your total points are equivalent to an 88%, you are guaranteed a B+ but a curve *may* move you into an A– .

Available points:

Quizzes: 100 points;
Exams: 200 points;
Final exam: 150 points
Total available: 450 points.

Extra credit: You will have opportunities during the term to receive extra credit points - to be added on top of your regular grade. **The maximum possible extra credit is 9 points.** Note: Extra credit will not turn a failing grade into an A.

Grade	A	A-	B+	B	B-	C+	C	C-	D+	D	D-	F
Points	≥ 418.5	≥ 405	≥ 391.5	≥ 373.5	≥ 360	≥ 346.5	≥ 328.5	≥ 315	≥ 301.5	≥ 283.5	≥ 270	< 270
Percent	93.0-100	90.0-92.9	87.0-89.9	83.0-86.9	80.0-82.9	77.0-79.9	73.0-76.9	70.0-72.9	67.0-69.9	63.0-66.9	60.0-62.9	Below 60.0

We do not negotiate grades. Feel free to ask questions about any grades you receive and to question if I made a mistake in the calculation of your grade. Please do not contact me once final grades are submitted to ask me to give you special treatment to raise your grade, i.e. “But I was so close to the higher grade…” or “I worked so hard...” are not acceptable requests, arguments, etc.

Your grades will be recorded on Blackboard.

Office for Access and Disability Services (ADS) and Learning Support Services (LSS):

It is the policy of Villanova to make reasonable academic accommodations for qualified individuals with disabilities. All students who need accommodations should go to Clockwork for Students via myNOVA to complete the Online Intake or to send accommodation letters to professors. Go to the LSS website http://learningsupportservices.villanova.edu or the ADS website https://www1.villanova.edu/university/student-life/ods.html for registration guidelines and instructions. If you have any questions, please contact LSS at 610-519-5176 or learning.support.services@villanova.edu, or ADS at 610-519-3209 or ods@villanova.edu.

Academic Integrity:

All students are expected to uphold Villanova’s Academic Integrity Policy and Code. Any incident of academic dishonesty will be reported to the Dean of the College of Liberal Arts and Sciences for disciplinary action. You may view the University’s Academic Integrity Policy and Code for a detailed description.

If a student is found responsible for an academic integrity violation, which results in a grade penalty, they may not WX the course unless they are approved to WX for significant medical reasons. Students applying for a WX based on significant medical reasons must submit documentation and their request for an exception will be considered.

Absences for Religious Holidays:

Villanova University makes every reasonable effort to allow members of the community to observe their religious holidays, consistent with the University’s obligations, responsibilities, and policies. Students who expect to miss a class or assignment due to the observance of a religious holiday should discuss the matter with their professors as soon as possible, normally at least two weeks in advance. Absence from classes or examinations for religious reasons does not relieve students from responsibility for any part of the course work required during the absence.
https://www1.villanova.edu/villanova/provost/resources/student/policies/religiousholiday s.html

Personal day[1]:

In addition to the attendance policy stated above, students are entitled to two excused absences for any reason that may contribute to their personal wellness. Students must advise the instructor by email before class of their intent to utilize a Personal Day as the reason for their absence. A Personal Day will not be approved retroactively. Students may, but are not required, to provide additional information regarding their absence. Additionally, Personal Days may not:

[1] Personal Days may not be used for the following: Labs, Clinicals, Internships, Courses that meet one time per week, Fast Forward courses offered by the College of Professional Studies, the online RN to BSN program, Summer Sessions, or graduate/law courses.

For more information: Villanova University’s Class Attendance Policy

- be used on consecutive class days;
- be used in the same week;
- be used immediately preceding or following a university holiday or break period;
- be used on days when exams, presentations or other major assignments are scheduled.

A Personal Day does not grant an automatic extension for items due. Students remain responsible for all assignments, exams, presentations, etc. due on that date. It is in the instructor's discretion to determine whether any extension is appropriate given individual circumstances.

Tentative Course Schedule F25:

Week	Dates	Agenda of topics	Quiz during recitation	Problems
1	**Aug 25-29**	Intro., Vectors, 1D Motion	R1 / Quiz 0 (Syllabus Feedback)	Chapter 1
2	**Sep 2 - 5**	1D motion	R2 / Quiz1 (Chapter 1)	Chapter 2
3	**Sep 8-12**	Multi-dimensional motion	R3 / Quiz2 (Chapter 2)	Chapter 3
4	**Sep 15 - 19**	Newton's First and Second Law, Forces and Free-body diagrams	R4 / Quiz3 (Chapter 3)	Chapter 4
5	**Sep 22 - 26**	Applications of Newton's law, Circular motion	R5 / Quiz4 (Chapter 4)	Chapter 5
6	**Sep 29 - Oct 3**	Gravity, **Review for Exam 1**	R6 / Quiz5 (Chapter 5)	Chapter 6
7	**Oct 6 - 10**	**Exam #1 (Chapters 1-5),** Kin. Energy, Work, Potential Energy	**R7 no quiz**	Chapter 7
8	**Oct 13 - 17**	**Fall Break (no classes)**	**Fall break**	
9	**Oct 20 - 24**	Work, Potential Energy, Energy Conservation	R8 / Quiz6 (Chapter 4,5)	Chapter 7,8
10	**Oct 27 - 31**	Center of Mass, Linear momentum, Elastic Collisions, Inelastic Collisions	R9 / Quiz7 (Chapter 7,8)	Chapter 9
11	**Nov 3 - 7**	Rotation, torque, Moment of inertia, Ang. momentum, Static equilibrium	R10 / Quiz8 (Chapter 9,10)	Chapter 10,11
12	**Nov 10 - 14**	Pressure, Archimedes principle**, Review for Exam 2**	R11 /Quiz9 (Chapter 11,12)	Chapter 12
13	**Nov 17 - 21**	**Exam #2 (Chapters 7-12),** Fluid statics	**R12 no quiz**	
14	**Nov 24 - 28**	Fluid dynamics (MT), **Thanksgiving break (WThF)**	**R13 no quiz**	Chapter 13
15	**Dec 1 - Dec 5**	Fluid dynamics, Oscillations SHM	R14 /Quiz10 (Chapter 13)	Chapter 14
16	**Dec 8 - Dec 11**	Waves and Sound, **Review for Final**	R15	Chapter 15,16
17	**Dec 15-Dec 19**	**Final Exam**		

Problem Assignments:

CH 1 – **P**: 1, 3, 6, 15, 21, 23, 31, 54, 56, 77, 78
CH 2 – **Q**: 2, 6. **P**: 1, 9, 17, 21, 27, 49, 54, 57, 66, 69, 91, 99
CH 3 – **Q**: 1,4,8,9,12,16,17,19. **P**: 2, 3, 7, 32, 35, 42, 63, 83,94
CH 4 – **Q**: 3,4,5,9. **P**: 1, 4, 45, 55, 56, 58, 59, 76, 90
CH 5 – **Q**: 7,14. **P**: 1, 19, 20, 21, 23, 28, 35, 40,46, 60
CH 6 – **P**: 3, 6, 8, 16, 62
CH 7 – **P**: 1, 6, 14, 17, 23, 38, 54, 58, 62, 93, 100
CH 8 – **P**: 1, 3, 5, 11, 14, 19, 24, 32
CH 9 – **P**: 2, 6, 8, 16, 21, 24, 32, 39, 44, 56, 58, 60
CH 10 – **P**: 6, 10, 23, 26, 27, 29, 32, 35, 40, 50, 54, 58, 63, 70, 71, 89
CH 11 – **P**: 1, 4, 22, 34, 40, 41, 49, 76
CH 12 – **P**: 2, 3, 6, 10, 14, 17, 21, 31, 42, 49
CH 13 – **P**: 10, 15, 16, 19, 27, 28, 33, 37, 41, 44, 45, 53, 57, 70, 78
CH 14 – **P**: 2, 4, 11, 16, 22, 29, 31, 40, 45
CH 15 – **P**: 1, 2, 7, 10, 15, 16, 23, 33, 46, 64
CH 16 – **P**: 1, 2, 11, 28, 47, 57, 58, 63, 64

Written Problem Procedure

1. Organize the problem.
    - Write out knowns and unknowns if possible.
    - Think about what you are trying to solve for.
    - Draw a diagram, if relevant
2. Determine how you are going to solve the problem.
    - Always write out the full equations before plugging in any values.
3. Solve and manipulate equations.
    - Try as often as possible to solve using variables only. Often variables cancel out!
    - Lay out your work in a manner that is easy to follow. I will take off points if I cannot read your solution.
    - Plug in numerical values
    - DO NOT FORGET UNITS AND DIRECTIONS (if a vector)
    - Circle/box your answer
4. Think about your answer. Does your answer seem reasonable?
    - You do not need to write anything for this. Yet, it helps to organize your thoughts.
    - Think about your value. Does it seem about the right order of magnitude? Does the sign (+/-) make sense?